

Acoustic impedance extraction method and acoustic characteristics analysis of perforated plates under grazing flow

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Abstract

As an important component of inlet and exhaust mufflers, the acoustic characteristics of perforated components are inevitably affected by the flow of air. Therefore, obtaining the acoustic impedance of the perforated element under airflow conditions is a prerequisite for accurate calculation of the muffler's muffling performance. In this work, the frequency domain linear Navier–Stokes (L-NS) method is used to extract the acoustic impedance of perforated plates under grazing flow. The predicted perforated acoustic impedance is consistent with the calculation results of published acoustic impedance expression, and the impedance boundary condition is defined to calculate the transmission loss (TL) of the perforated muffler, which agrees well with the experimental results and verifies the accuracy of the method. The effect of perforation angles on the transmission loss of mufflers in different Mach numbers (Ma) and aperture plate thickness ratio (d_h/t_p) is analyzed by the frequency domain L-NS method. The results show that when $1 \leq d_h/t_p < 2$ and $Ma \leq 2$, the effect of perforation angles on the muffler performance is obvious, and the angle tilted upstream shifts the resonant frequency to a lower frequency while its corresponding peak value is also increased. As an engineering application, it has certain significance for the prediction of muffler muffling performance and the regulation of the muffling frequency band.

Keywords

Grazing flow, perforated plate, acoustic impedance, frequency L-NS method, perforation angle, transmission loss

Introduction

Perforated elements are widely used in inlet and exhaust mufflers^{1–3} and combustion systems^{4,5} to improve the muffling performance in a specific frequency range, reduce flow resistance¹ and prevent thermoacoustic oscillations.^{6,7} The application context of this paper is to analyse the effect of grazing over-flow on the muffling performance of perforated mufflers at room temperature, so the acoustic impedance of perforated elements^{8–10} is necessarily influenced by the flow factors near the perforated plate and the resulting eddy currents^{7,8} are the main mechanism affecting sound dissipation.

In recent decades, the perforated acoustic impedance under grazing flow has attracted the attention of many scholars. The research methods mainly include: theoretical analysis, experimental measurements and numerical calculations. Howe¹¹ theoretically investigated the effect of swept flow on the acoustic conductivity of small holes and gave an analytical solution for the acoustic conductivity, but made more assumptions in the derivation process, resulting in a large difference from the experimental values. Jing¹² et al. used the relevant theory of Howe¹¹ and used the displacement continuum condition to study the acoustic impedance of perforations at a specific Mach number. The theoretical calculations were in good agreement with the experimental measurements, but the distribution of acoustic impedance with frequency was not given. Based on Jing,¹² Peat¹³ summarised the variation of acoustic impedance with the Strouhal number (S_r) under the action of grazing flow by setting up displacement continuity conditions and velocity continuity conditions, it is not reasonable that

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the perforated acoustic impedance as a whole but requires different continuity conditions. Recently, Meng¹⁴ derived a semi-empirical formulation for perforation using a linearised potential flow model, but the model predictions only agreed well with experimental measurements at low S_r . Zhao¹⁵ developed a one-dimensional theoretical model of the pneumatic-acoustic damping effect of a perforated plate in the case of biased flow by modifying the Cummings equation for non-constant flow with the Cargill equation for acoustic open boundary conditions at the end of the pipe. Guan^{16,17} studied the acoustic damping performance of single and double coaxial perforated plates and double off-axis perforated plates at low Mach number and Helmholtz number based on acoustic theory and experiments, and summarised the influence of structural parameters of double perforated plates on their acoustic damping performance, but there are some differences between the theoretical model and experimental values of double perforated plates, mainly because it is difficult to consider the vortex effect. Yang¹⁸ developed an impedance model for the lining of elongated pipes under the action of grazing flow based on the transfer matrix method, which provides better accuracy in predicting transmission loss. In practice, mathematical models that can fully account for the effect of flow-acoustic coupling on the acoustic impedance of grazing flow are difficult to develop. Rao,¹⁹ Gummings²⁰ and Kirby²¹ experimentally measured the acoustic impedance of perforations under grazing flow conditions and obtained a variety of empirical formulae for acoustic impedance, but different impedance extraction strategies and flow boundary conditions led to significant differences in the acoustic impedance of small holes calculated by these empirical formulae. With the development of computer technology and simulation software, numerical methods are able to characterise the above conditions very well. Tam² calculated the effect of slit resonator acoustic performance by direct numerical modelling (DNS) to study the reflection coefficients of slits at different tilt angles, and their calculated results agreed well with experimental test values, demonstrating that DNS is a useful and accurate tool for linear aerodynamic acoustic prediction. Zhang³ calculated the interaction of the laminar and turbulent boundary layer states with the liner bore by DNS at different incident sound field parameters and with the laminar and turbulent boundary layer states, and showed that the bushing increases the drag with increasing incident sound pressure amplitude for all conditions studied. For practical applications, Zhong and Zhao²² developed a one-dimensional time-domain numerical model to simulate the propagation and dissipation of acoustic plane waves in a cylindrical lined pipe, and the results of the model agreed well with previous experimental and frequency domain results. Ji and Zhao²³ applied the lattice Boltzmann method (LBM) to study the acoustic damping effect of circular holes in pipes, which not only has higher accuracy but also improved computational efficiency compared with the conventional NS solver. Ji,²⁴ Liu²⁵ and Zhu²⁶ predicted the transmission loss of perforated muffler in the presence of airflow using the time-domain CFD method, and the calculated results agreed well with the experimental measurements, which confirms the accuracy of the time-domain CFD method, but the time-domain CFD method has low computational efficiency, so its application in the actual design of muffler is limited. As a result, some researchers^{27–29} have fitted acoustic impedance expressions by time-domain CFD methods to improve computational efficiency, Zhang²⁷ used the time-domain CFD method to study the linear relationship between the internal flow parameters and the flow-related terms in the acoustic impedance formula, and established the acoustic impedance expression within a certain flow range. Chen^{28,29} used the time-domain CFD method to study the effect of structural parameters and vortex velocity in the hole on the acoustic impedance, calculated the contribution of the perforated plate to the acoustic impedance of the bias-flow and grazing flow conditions, and fitted an exact expression for the perforated acoustic impedance, and the calculated muffler transmission loss agreed well with the experimental value in most cases, but because of the limitation of the extraction method, it is not suitable for the case above 0.2 Mach number. Therefore, this paper will use the L-NS equation to solve the perforation acoustic impedance in the frequency domain, Chu³⁰ and Ullrich³¹ proved that the L-NS equation can characterise the linear propagation process of acoustic wave, vortex wave and entropy wave, and the interaction between the three acoustic modes and the mean flow. Xin³² found that turbulent viscosity is an important effect on the coupling of flow and acoustics when solving the L-NS equation in the frequency domain. Zhao³³ and Huang³⁴ predicted the transmission loss of the muffler using the L-NS equation, and the calculated results were in good agreement with the experimental values. Toulorge³⁵ used the frequency domain L-NS equation to calculate the incident and radiated acoustic impedance of a square perforation at low flow rates, and studied the influence of the width, thickness and edge shape of the square perforation on the dimensionless acoustic impedance. However, Toulorge³⁵ selected a more complex acoustic impedance extraction method, the calculation method occupied a higher computer memory. Ou³⁶ developed the impedance boundary Navier–Stokes equation (IBNSE) method to predict the absorption characteristics of acoustic ducts with grazing flow.

In this work, the perforation acoustic impedance is calculated using a frequency domain approach under grazing flow. The main structure is as follows: The second section describes the theory and extraction method of the perforated acoustic impedance under grazing flow. The third section introduces the establishment and meshing of the perforation flow field model under grazing flow, analyzes the velocity distribution and shear rate near the perforation and maps the flow field information to the acoustic field model. The fourth section introduces the calculation method of the perforated acoustic

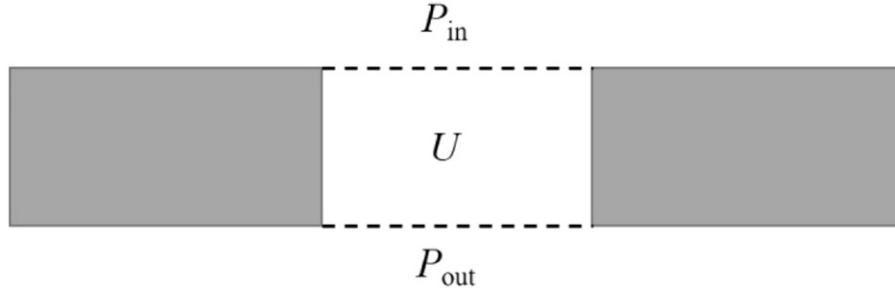


Figure 1. Acoustic quantities on both sides of perforation.

field model under grazing flow, compares the calculated perforated acoustic impedance with the theoretical value, and compares the transmission loss of the muffler with the experimental value. The fifth section summarises the effect of perforation angle on transmission loss under grazing flow.

The extraction method of perforation acoustic impedance

For the circular perforation shown in Figure 1, the acoustic impedance⁸ is defined as

$$Z_{trans} = \frac{\Delta P}{U} = \frac{P_{in} - P_{out}}{U} = R_0 + jX_0 = R_0 + jp_0ckl \quad (1)$$

In the formula, P_{in} and P_{out} are the sound pressure at the inlet and outlet of the perforation, respectively, and the volume velocity in the perforation, R_0 is the acoustic resistance, X_0 is the acoustic reactance, c is the speed of sound, k is the wave number, j is the imaginary unit and l is the equivalent depth of the circular hole. Therefore, the difference between the sound pressure on both sides of the perforation and the ratio of the volume velocity through the perforation is the acoustic impedance of the perforation, as can be seen from formula (1), the acoustic impedance of the perforation can be calculated as long as the pressure drop on both sides of the perforation and the volume velocity in the perforation are obtained.

To obtain the sound pressure at the inlet and outlet of the perforated plate and the volume velocity in the hole under grazing flow in COMSOL, it is necessary to establish a suitable model of the flow field and the acoustic field, and the non-local coupled integration of the sound pressure in the sound field and the boundary where the volume velocity of the central section of the perforation is located. The volume velocity in the perforation is expressed as $U = \int u \cdot n dA$. As shown in Figure 2, the acoustic model selected cross-sections of the perforated plate inlet and outlet with the central boundary selected in the hole. In this section, a single-hole model is established. If a multi-hole model is established in the flow and sound acoustic model, it not only increases the calculation cost of the flow field and the acoustic field, but also decreases the accuracy of the calculation results due to the decrease in mesh quality caused by the complexity of the structure.

Among them, r is the perforation radius, t_p is the plate thickness and L_x and L_y are the x -direction spacing and y -direction spacing of the centre line between the perforations, respectively. The plane wave is incident from the perforated inlet section and simulated by the ‘background sound field’ feature in the ‘linear Navier-Stokes, frequency’ interface. When extracting in the three-dimensional frequency domain L-NS, it is necessary to define the perforation acoustic impedance extraction formula, that is

$$\Delta P = \frac{(\text{intop_in}(\text{lnsf.p.t}) - \text{intop_out}(\text{lnsf.p.t}))}{\text{intop_in}(1)} \quad (2)$$

$$U = \text{intop_mid}(u) \quad (3)$$

$$Z_{trans} = \frac{\Delta P \cdot \text{intop_in}(1)}{U} \quad (4)$$

In the formula, lnsf.p.t is the sound pressure expression in the ‘linear Navier-Stokes, frequency’ interface, $\text{intop_in}(\text{lnsf.p.t})$ is the sound pressure of the inlet boundary, $\text{intop_out}(\text{lnsf.p.t})$ is the sound pressure of the outlet

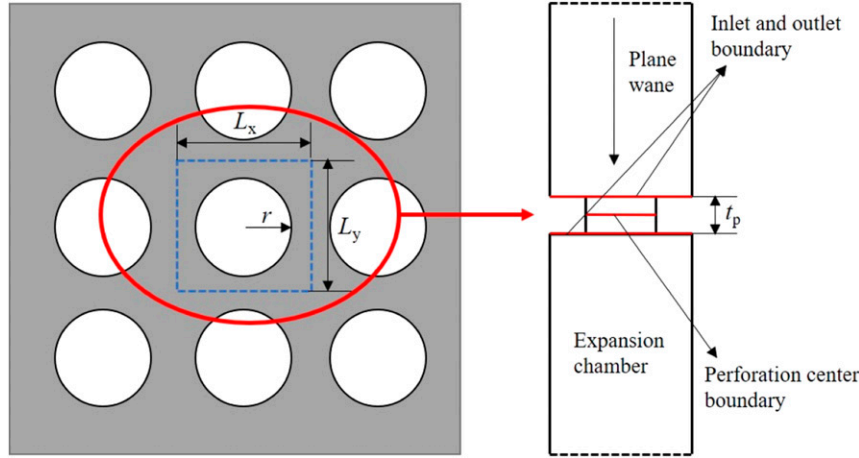


Figure 2. The boundary of inlet, outlet and hole of perforated plate.

boundary, $\text{intop_in}(1)$ is the unit area of the inlet boundary and $\text{intop_mid}(u)$ is the volume velocity of the perforation centre boundary.

Flow characteristics analysis and information mapping

Computation module

In this section, the velocity distribution and shear rate distribution at the perforation are analyzed in detail. The structural parameters of the perforated plate used in the section are as follows: perforation diameter $d_h = 6$ mm, plate thickness $t_p = 2$ mm and porosity $\varphi = 15\%$.

The flow field calculation model includes the main pipe, perforations and side branch pipes, the perforations are placed at the bottom of the main pipe, the airflow flows through the upper side of the perforations, the flow medium is an ideal gas at room temperature, the stress tensor and linearised equation of state³⁷ is expressed as

$$\sigma = -pI + \mu[\nabla u + (\nabla u)^T] + \left(\mu_B - \frac{2}{3}\mu\right)(\nabla \cdot u)I \quad (5)$$

$$\rho = \rho_0(\beta_T p - \alpha_p T) \quad (6)$$

where p , $\mathbf{u}$, T and ρ are the pressure, velocity, temperature and density perturbations, respectively, μ and μ_B are the dynamic and bulk viscosity of the material, respectively, $\mathbf{I}$ is the unit matrix, α_p and β_T are the isobaric thermal expansion and isothermal compression coefficients, respectively.

The upstream pipe of the main is 150 mm long and the inlet is set to fully developed flow to allow the turbulent boundary layer to fully develop and the downstream pipe is 50 mm to allow the shedding vortex to be fully depleted. The fluid calculations are modelled using SST RANS in the CFD module of COMSOL, which has a greater advantage in solving for free shear flow and boundary layers, and the velocity boundary conditions for the inlet can be written as

$$U_{\text{in}} = Ma \cdot c_0 \quad (7)$$

where Ma denotes the Mach number of the airflow.

In order to skim the effect of over-flow on the acoustic impedance of the perforation, it is necessary to define the background mean flow temperature T_0 , the absolute pressure p_0 and the velocity field $\mathbf{u}_0$. In this paper, T_0 is taken to be 293.15 K, while the absolute pressure and velocity fields are obtained from the turbulent physical field. It is assumed that there is no tangential stress at the boundary, so the following expression for the boundary conditions holds

$$\mathbf{n} \cdot \mathbf{u} = 0, \sigma_n - (\sigma_n \cdot \mathbf{n})\mathbf{n} = 0, \sigma_n = \sigma \mathbf{n} \quad (8)$$

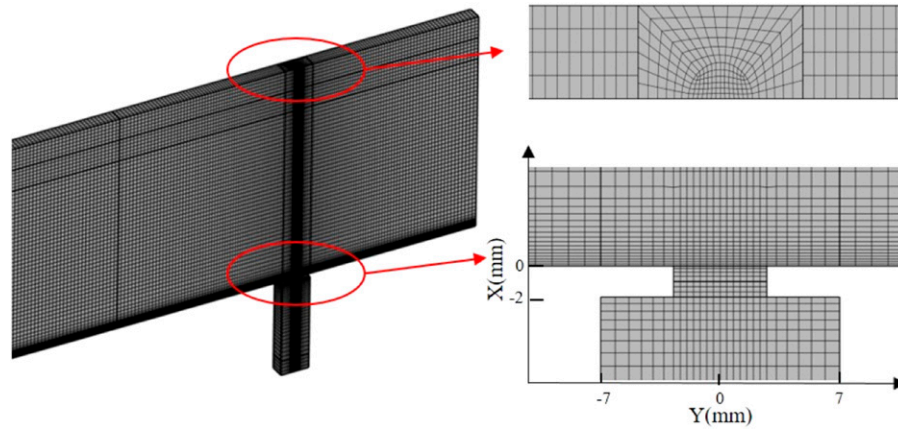


Figure 3. Flow field grid model.

where $\mathbf{n}$ is the normal unit vector.

The use of hexahedral structural grids can obtain accurate calculation results and reduce calculation time. Figure 3 shows the flow field grid model of the perforated plate acoustic impedance calculation model. The mesh size of the main pipeline is 1 ~ 2 mm. The thickness of the first layer of the boundary layer on the upper side of the perforation is divided according to the principle of $y^+ \approx 2$. The thickness is between 0.05 ~ 0.1 mm, and the boundary layer is at least 15 layers. The minimum mesh size in the X -axis direction is 0.07 mm, and the mesh size in the Y -axis direction is 0.33 mm. Due to the symmetry of the perforated model, symmetric surface boundary conditions can be used to reduce the degrees of freedom of the calculation in order to improve computational efficiency. Based on the above meshing principles, the total number of meshes is 103,334. The following comparison with the experimental data²⁹ from the references demonstrates that the number of meshes is sufficient.

Flow field information at the perforated plate

This section investigates the gas flow phenomenon at the perforation in terms of both the surface velocity and shear rate on the perforation, respectively. Figure 4 shows a cloud of the steady state flow velocity distribution near the perforated plate for a grazing flow Mach number of 0.1. It can be seen from the diagram that almost all of the air flows sideways from the perforated plate, forming a velocity buffer zone in the downstream region of the perforation, while a small amount of gas enters the side branch pipe through the perforation, and the airflow in the side branch pipe is very small. As shown in Figure 5, the airflow forms a clearly layered shear layer on the upper surface of the perforation, which has a large effect on the perforation acoustic impedance due to the perturbing effect of the shear layer and the acoustic waves. To study the shear layer flow phenomenon near the perforated plate in detail, the distribution of y -velocity in y -direction is studied at the line segment AB on the upper surface of the perforation. According to the research of some scholars, the eddy current is usually distributed in the shear layer of the upper surface of the perforation. The velocity of the eddy current is the fluid velocity in the shear layer of the upper surface of the perforation, which gradually increases with the increase of the grazing flow Mach number. (Figure 6)

Flow field information mapping

The flow field information is mapped to the acoustic mesh by multi-physics field coupling³⁸ in COMSOL, and the solution of the flow field information in the CFD mesh is compared with the mapped solution on the acoustic mesh below. A three-dimensional section line is selected along the perforation centre in the y -direction of the main pipeline. The mapping results on the section line are transformed into a one-dimensional chart form, which can intuitively judge the accuracy of the mapping results. Figure 7 shows the comparison between the axial (cross-section direction) internal velocity calculated on the CFD grid and the mapping results on the acoustic grid. The flow field information solution is almost consistent with the mapping solution curve. In the flow field information, the turbulent viscosity is an important factor in the interaction between the reaction flow and the acoustics. Therefore, the map shows that the turbulent viscosity distribution calculated on

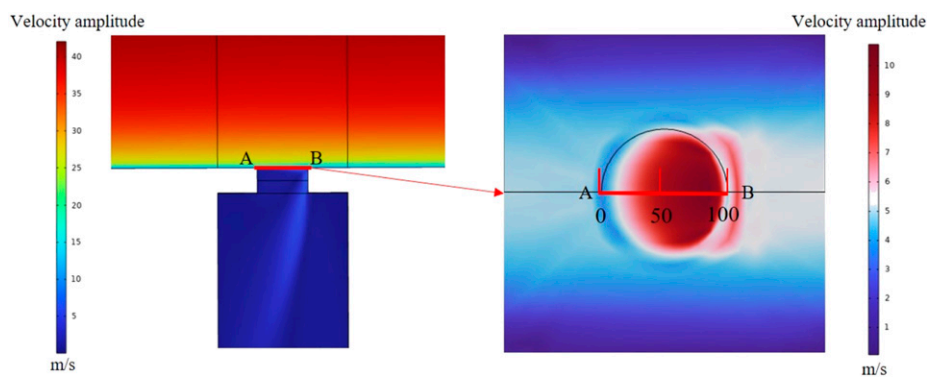


Figure 4. Velocity cloud near the perforated plate.

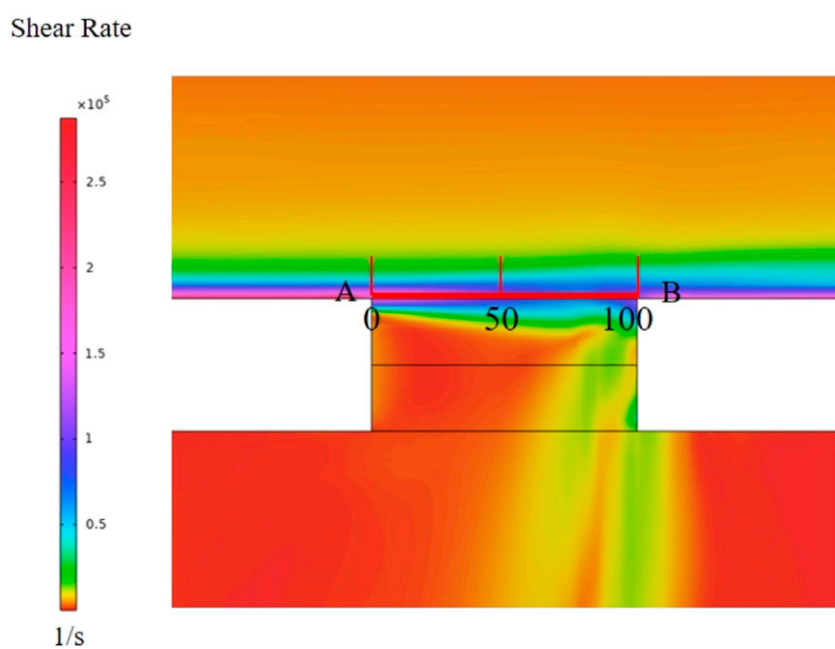


Figure 5. Distribution of shear rate near the perforated plate.

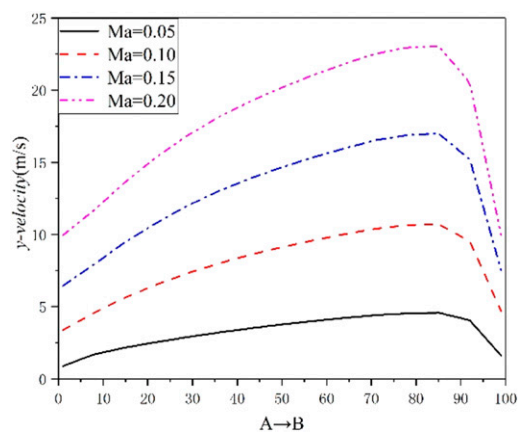


Figure 6. Velocity distribution on line segment AB at the perforation.

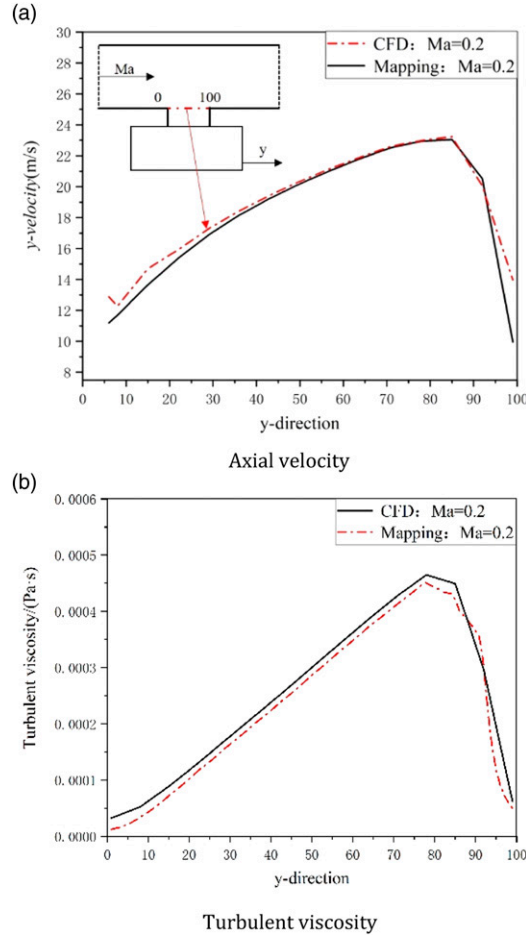


Figure 7. Comparison of mapping results. (a) Axial velocity; (b) Turbulent viscosity.

the CFD grid is compared with the mapping results on the acoustic grid. The change of the flow field information solution and the mapping solution curve is almost the same.

Sound field calculation and model verification

The equations solved in acoustic calculations are the linearised Navier–Stokes equations³⁹ given in the frequency domain, as follows

$$i\omega\rho + \nabla \cdot (\rho_0\mathbf{u} + \rho\mathbf{u}_0) = M \quad (9)$$

$$\rho_0[i\omega\mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u}_0 + (\mathbf{u}_0 \cdot \nabla)\mathbf{u}] + \rho(\mathbf{u}_0 \cdot \nabla)\mathbf{u}_0 = \nabla \cdot \boldsymbol{\sigma} + \mathbf{F} - \mathbf{u}_0M \quad (10)$$

$$\rho_0c_p[i\omega T + \mathbf{u} \cdot \nabla T_0 + \mathbf{u}_0 \cdot \nabla T] + (\rho c_p)\mathbf{u}_0 \cdot \nabla T - \alpha_p T_0[i\omega p + \mathbf{u} \cdot \nabla p_0 + \mathbf{u}_0 \cdot \nabla p] - (\alpha_p T)\mathbf{u}_0 \cdot \nabla p_0 = \nabla \cdot (\kappa \nabla T) + \Phi + Q \quad (11)$$

$$\rho = \rho_0(\beta_T p - \alpha_p T) \quad (12)$$

where the mass M , force $\mathbf{F}$ and heat source term Q are zero, and p_0 , $\mathbf{u}_0$, T_0 and ρ_0 density refer to the steady state background pressure, velocity, temperature and density, respectively. ω is the angular frequency of the incident wave, k is the wave number and σ is the stress tensor.

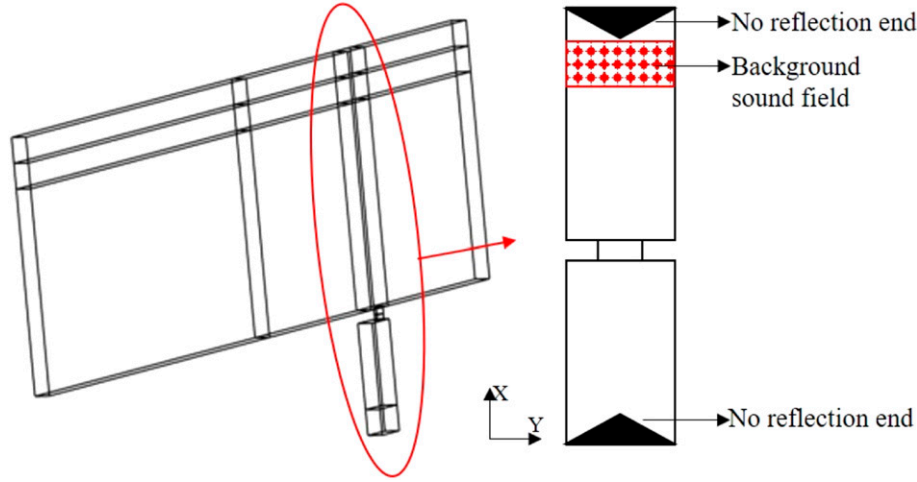


Figure 8. Acoustic calculation model.

The acoustic calculations are made in 10Hz frequency steps and include all of the following. The acoustic calculation model does not require the overall flow field model. As shown in Figure 8, a part of the main pipeline is intercepted with the size of the side branch pipeline as the constraint. The flow field information is automatically mapped to the interception part. A non-reflective end is added at the end of the interception model to avoid the reflection of sound waves in the pipeline and affect the accuracy of the extraction results. A background sound field is applied near the end of the perforation to generate a plane wave signal. The plane wave variable^{40,41} is defined as follows

$$P_b = 1\text{Pa} \cdot e^{-ik_0(-x)} \quad (13)$$

$$k_0 = \frac{\omega}{c_0 + U_{in}} \quad (14)$$

$$v_b = \left(0, 0, -\frac{1}{i\omega\rho_0} \frac{\partial P_b}{\partial x}\right) \quad (15)$$

In the formula, P_b is the sound pressure of the background sound field; k_0 is the wave number in the air; v_b is the background sound velocity vector; c_0 is the sound velocity in the air; U_{in} is the grazing velocity; ρ is the air density. In the formula, lnsf.p.t is the sound pressure expression in the 'linear Navier-Stokes, frequency' interface, $\text{intop_in}(\text{lnsf.p.t})$ is the sound pressure of the inlet boundary, $\text{intop_out}(\text{lnsf.p.t})$ is the sound pressure of the outlet boundary, $\text{intop_in}(1)$ is the unit area of the inlet boundary and $\text{intop_mid}(u)$ is the volume velocity of the perforation centre boundary.

To verify the accuracy of the acoustic impedance extracted by the L-NS equation in the frequency domain, the calculated results are compared with the acoustic impedance fitting equation given by Chen²⁹ under grazing flow. Figure 9 shows the curve of acoustic impedance and acoustic impedance changing with the Strohal number S_t for Mach number is 0.1 and 0.2. It can be seen from the figure that when S_t is small, the grazing flow increases the acoustic resistance of the perforation and the acoustic energy absorbed by the perforated plate increases, and when S_t exceeds 0.7, the acoustic resistance drops to a negative value, indicating that flow noise is generated at the small perforation; when S_t is small, the grazing flow reduces the acoustic resistance of the perforation, and as S_t increases, the acoustic resistance increases sharply. The model of perforated plate (L-NS) and Chen²⁹ calculate the results of the change of trend and numerical match well, but there is a numerical error at high S_t , which is acceptable. The following will indirectly prove the accuracy of the model of this paper (L-NS) by calculating the transmission loss of perforated muffler. (Table 1)

To further verify the accuracy of the perforation acoustic impedance extraction method in this study, the transmission loss of the perforated pipe muffler will be calculated by applying the obtained acoustic impedance to the perforated plate through the finite element method and comparing it with the experimental test value. (Figure 10)

Figure 11 shows the model (L-NS) and Chen²⁹ compared to experimental measurements for Mach numbers of 0.1 and 0.2. The muffler M1 has a small aspect ratio, and the transmission loss curve shows only one resonance peak in the range of

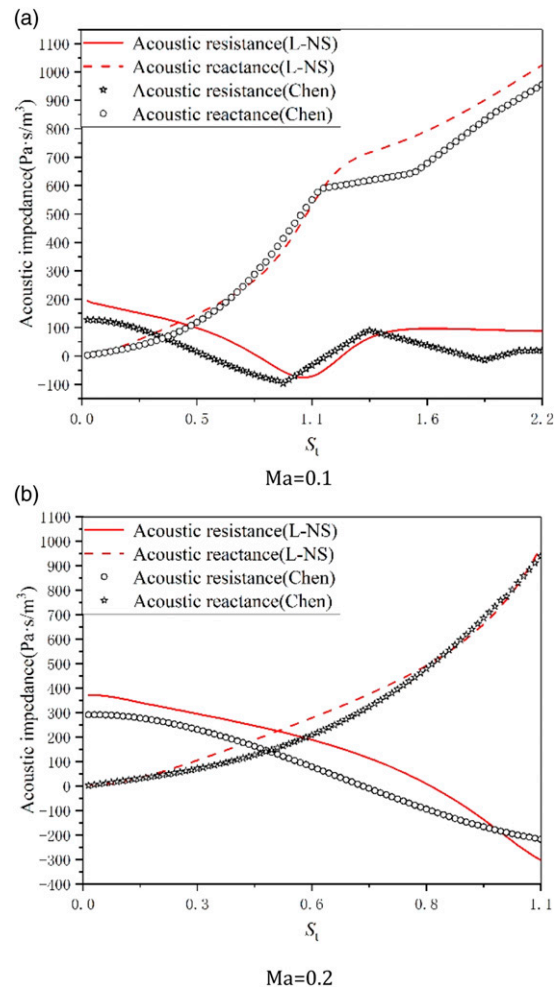


Figure 9. Comparison of acoustic impedance and acoustic reactance. (a) $Ma = 0.1$; (b) $Ma = 0.2$.

Table 1. Structure size parameters of perforated tube muffler (unit: mm).

Geometric parameter	Sign	MI
Length	l	40
Diameter	D	110
Perforated tube diameter	d	32
Perforation diameter	d_h	6
Perforated tube thickness	t_p	2
Perforation percentage	φ	15%
Mach number	Ma	0.1, 0.2

0–4000Hz. As can be seen from Figure 11(a), the model agrees well with the experimental values of Chen,¹⁸ but at a Mach number of 0.2, there is a deviation in the resonance frequency between the two models, which may be caused by the high acoustic impedance of Chen⁰. It can be seen that the transmission loss obtained from the acoustic impedance of the model in this paper agrees well with the experimentally measured values in most cases, thus verifying the accuracy of the model in this paper.

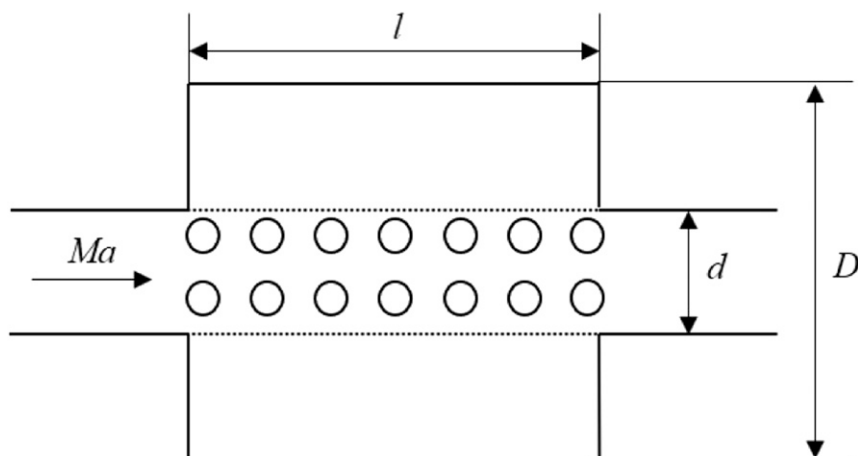


Figure 10. Structure diagram of the perforated muffler.

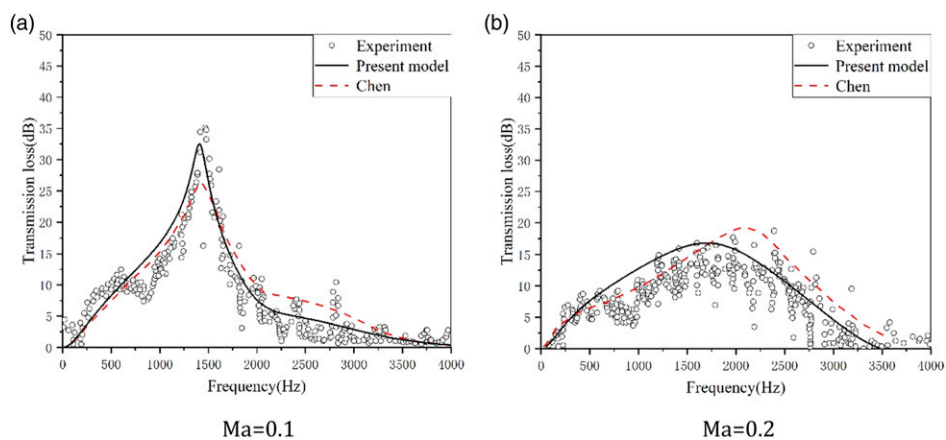


Figure 11. The transmission loss of down muffler under swept flow. (a) $Ma = 0.1$; (b) $Ma = 0.2$.

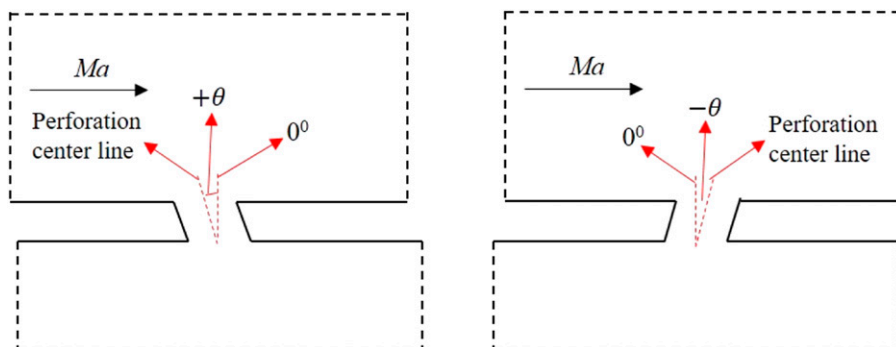


Figure 12. Perforation angle model diagram.

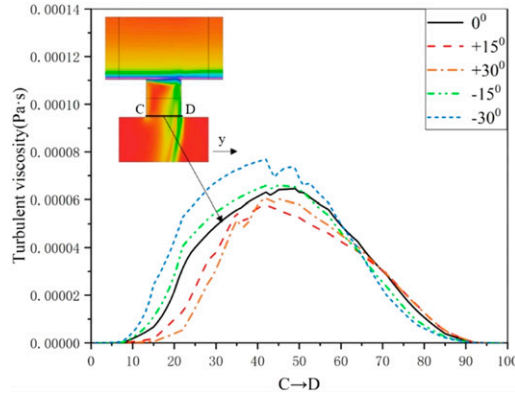


Figure 13. Turbulent viscosity distribution.

Effect of perforation angle on transmission loss

According to Zhao,⁴² it was found that the geometry of the perforation has a non-negligible effect on the acoustic performance of perforated plates in the flow state. This paper will extend to the effect of perforation angle on the acoustic impedance of perforations under the action of swept flow. The perforation angle is defined as the angle between the central axis of the perforation and the vertical direction, As shown in Figure 12, the tilt of the axis upward is denoted as ‘ $+\theta$ ’, and the tilt of the axis downward is denoted as ‘ $-\theta$ ’. The effect of the perforation angle on the acoustic resistance R_0° and acoustic reactance X_0° of the vertical perforation (0°) is investigated by numerical calculations at different grazing flow Mach numbers Ma

$$r_0 = \frac{(R_\theta - R_{0^\circ})}{\rho c} \quad (16)$$

$$x_0 = \frac{(X_\theta - X_{0^\circ})}{\rho c} \quad (17)$$

where R_θ and X_θ are the sound resistance and sound reactance for the corresponding perforation angles, r_0 and x_0 are the dimensionless contributions of the corresponding perforation angles to the vertical perforation, ρ is the air density and c is the sound velocity in air.

Effect of perforation angle on transmission loss at different Ma

The size of the muffler model used in this section is $l = 50$ mm, $D = 90$ mm, $d_h = 3$ mm, $t_p = 3$ mm, $\phi = 12\%$ and $Ma = 0.05, 0.1, 0.2$. To study the flow characteristics of the keyhole at different perforation angles, the distribution of turbulent viscosity in the y direction is studied at the line segment CD of the lower surface of the keyhole. Figure 13 shows the turbulent viscosity of the transverse line at the centre of the perforation outlet at different perforation angles when the Mach number is 0.1. The abscissa is the average distribution point in the y direction from the inner wall near the incoming flow direction as the starting point (point C) to the endpoint (D). As shown in Figure 13, the larger the angle of inclination from upstream to downstream, the more the turbulent viscosity profile expands towards the inner wall in the incoming direction, and the stronger the spread of turbulent vortex clusters, and the expansion becomes more pronounced as the Mach number increases, indicating that the angle of inclination to the downstream under the effect of the swept flow will enhance the sound perturbation effect of the flow, which may lead to the deterioration of the muffler’s muffling performance.

Figure 14 shows the effect of perforation angle on acoustic impedance at different Mach numbers. It can be seen from the figure that the dimensionless acoustics resistance r_0 is positive when the perforation angle is tilted downstream, and the positive value becomes larger as S_t and Mach number increase, indicating that the angle tilted upstream enhances the acoustics resistance r_0 , resulting in enhanced sound absorption performance of the perforated plate; The dimensionless acoustics reactance x_0 is positive and increases as S_t rises, indicating that the angle of inclination upstream will weaken the acoustics reactance x_0 , resulting in more sound energy being radiated by the perforated plate. When the perforation angle is tilted upstream, the dimensionless acoustics resistance r_0 is negative and increases with S_t and Mach number, indicating that

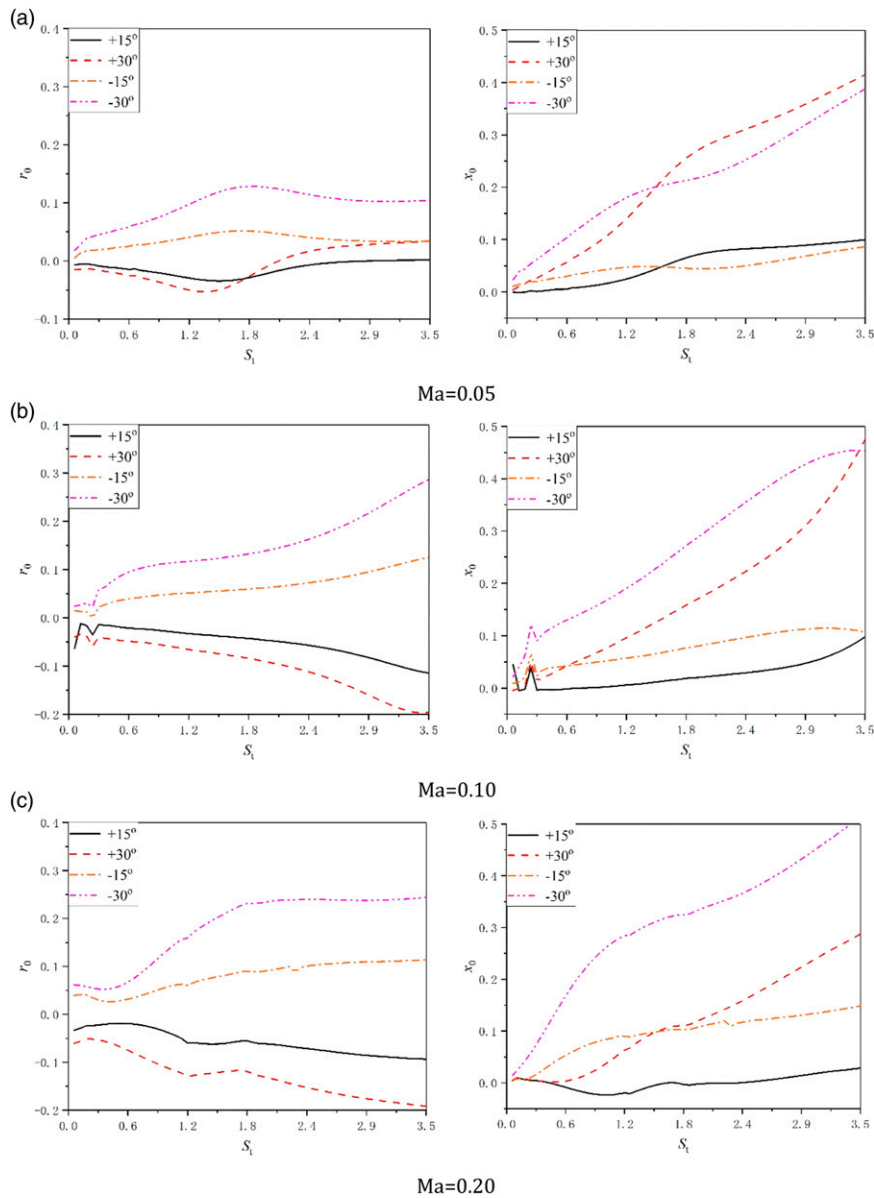


Figure 14. Effect of perforation angle on acoustic impedance at different Mach numbers. (a) $Ma = 0.05$; (b) $Ma = 0.10$; (c) $Ma = 0.20$.

the angle tilted downstream weakens the acoustics resistance r_0 and leads to a weakening of the perforated panel's sound absorption performance; The dimensionless acoustics reactance x_0 is positive and increases as S_1 rises, indicating that the angle of inclination downstream will enhance the acoustics resistance x_0 , resulting in more sound energy being radiated by the perforated plate. Overall, the degree of variation of the dimensionless acoustics reactance x_0 is greater than the dimensionless sound resistance r_0 , which may lead to the acoustics reactance x_0 being the main factor in the variation of perforated plate acoustic properties due to the angle of perforation.

Figure 15 shows the transmission loss comparison of each perforation angle at different Mach numbers. As shown in the figure, when the Mach number is 0.05, the larger the perforation angle, the higher the offset of the resonant frequency of the muffler to the lower frequency, and as the Mach number increases, the offset of the perforation angle tilted downstream is gradually larger than that tilted upstream, while the peak value of the transmission loss tilted upstream is obviously larger than that corresponding to the peak value tilted downstream, mainly because the tilted angle increases the equivalent thickness of the perforated plate and the effect of the grazing flow. The angle of inclination to the downstream will enhance the perturbing effect of the flow on the sound, enhancing the acoustics reactance value and weakening the muffling

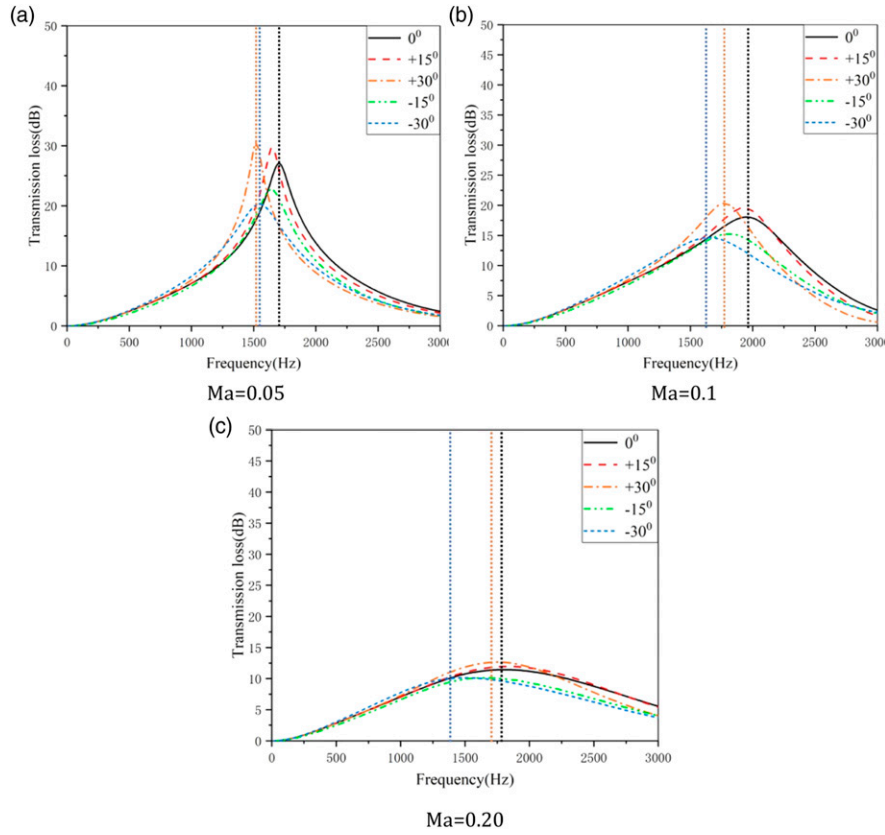


Figure 15. Effect of perforation angle on transmission loss at different Mach numbers. (a) Ma = 0.05; (b) Ma = 0.1; (c) Ma = 0.20.

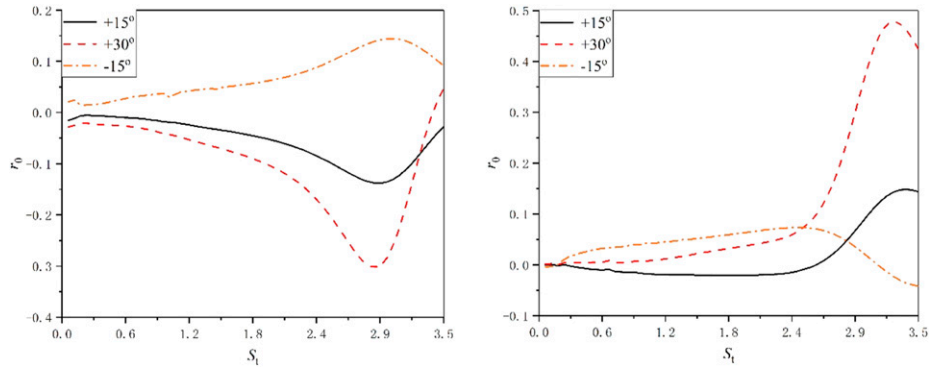


Figure 16. The effect of perforation angle on acoustic impedance for an aperture plate thickness ratio of two.

performance of the muffler, which is in line with the description in Figure 14. As the Mach number increases, the effect of the perforation angle on the transmission loss is limited, mainly because the structural micro-variations at high flow rates affect the anechoic performance to a limited extent and the transmission loss curve becomes increasingly smoother.

Effect of perforation angle on transmission loss at different d_h/t_p

The model used is a muffler with length $l = 50$ mm, diameter $D = 90$ mm, perforation rate $\phi = 12\%$, Mach number $Ma = 0.1$ and aperture plate thickness ratios of $d_h/t_p = 1, 1.5, 2$ and 3 . The main reason for not considering the scheme of $d_h/t_p > 2$ is

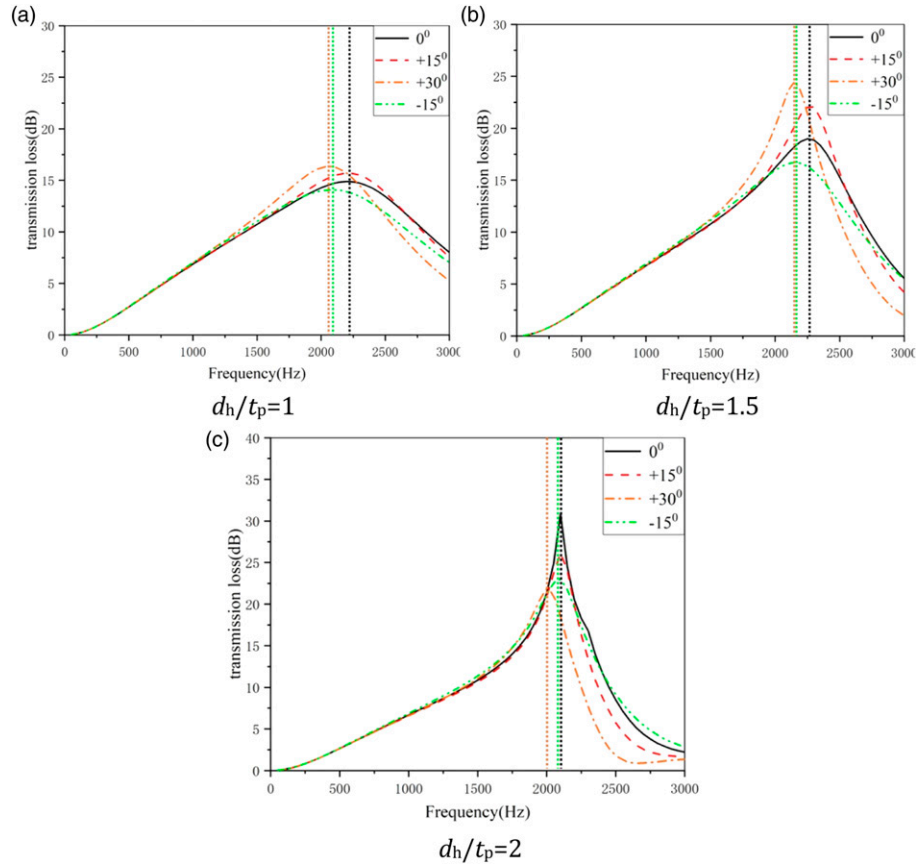


Figure 17. Effect of perforation angle on transmission loss at different aperture plate thickness ratios. (a) $d_h/t_p = 1$; (b) $d_h/t_p = 1.5$; (c) $d_h/t_p = 2$.

that the structural characteristics of perforation angle cannot be highlighted. Figure 16 shows the effect law of perforation angle on acoustic impedance when the aperture plate thickness ratio is 2. As can be seen from the figure, when S_t is small, the variation of dimensionless sound resistance and sound resistance is not great. When $S_t = 2.9$, the dimensionless acoustics resistance r_0 and acoustics reactance x_0 will show a trough and crest variation, which will cause the muffler's muffling performance to deteriorate at high frequencies.

As shown in the Figure 17, when $d_h/t_p = 1$ and 1.5, the effect of the perforation angle on the transmission loss of the muffler is more obvious, and the angle of the upstream tilt will still enhance the peak value of the resonance frequency. However, when $d_h/t_p = 2$, the existence of the perforation angle weakens the muffler performance, and the larger the perforation angle, the more obvious the weakening, this coincides with the depiction in Figure 16.

Discussion and conclusions

The purpose of this work is to obtain the accurate acoustic impedance of the perforated plate under grazing flow, to calculate the transmission loss of the muffler, and to investigate the effect of the perforation angle on the transmission loss of the muffler under different Mach number (Ma) and aperture plate thickness ratio (d_h/t_p). The main conclusions of the study are as follows:

1. The perforated plate acoustic impedance is extracted by defining the perforated plate inlet and outlet sound pressure and the volume velocity in the hole, respectively, in the frequency domain linear Navier–Stokes equation. The extracted perforated acoustic impedance is defined in the framework of the finite element method to calculate the transmission loss of the perforated muffler, and the calculation method has a high prediction accuracy in most cases by comparison with experimental data.

2. The presence of the perforation angle disguisedly increases the equivalent thickness of the perforated plate and acoustics reactance, shifting the resonant frequency of the muffler towards lower frequencies. Under grazing flow, when $0 < Ma \leq 2$, the peak value of the transmission loss tilted upstream is greater than that tilted downstream with vertical perforation, while the angle of perforation tilted downstream enhances the perturbation effect of flow on acoustics, leading to an increase in sound reactance and a decrease in the muffler's muffling performance. When $Ma > 2$, the presence of perforation angle has less effect on the transmission loss.
3. Under grazing flow, when $1 \leq d_h/t_p < 2$, the angle of inclination upstream still has a relatively positive effect on the transmission loss, but when $d_h/t_p \geq 2$, the presence of the perforation angle weakens the muffler's muffling performance, and the larger the perforation angle, the more obvious the weakening.

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References

1. Munjal ML. *Acoustics of ducts and mufflers with application to exhaust and ventilation system design*. Hoboken, New Jersey: John Wiley & Sons, 1987.
2. Tam CKW, Ju H, Jones MG, et al. A computational and experimental study of slit resonators. *J Sound Vib* 2005; 284(3–5): 947–984.
3. Zhang Q and Bodony DJ. Numerical investigation of a honeycomb liner grazed by laminar and turbulent boundary layers. *J Fluid Mech* 2016; 792: 936–980.
4. Lawn C. The acoustic impedance of perforated plates under various flow conditions relating to combustion chamber liners. *Appl Acoust* 2016; 106: 144–154.
5. Cai T, Zhao D, Sun Y, et al. Evaluation of NOx emissions characteristics in a CO₂-Free micro-power system by implementing a perforated plate. *Renew sustain energy rev* 2021; 145: 111150.
6. Zhao D, Gutmark E, and Reinecke A. Mitigating self-excited flame pulsating and thermoacoustic oscillations using perforated liners. *Sci Bull* 2019; 64(13): 941–952.
7. Zhao D and Li J. Feedback control of combustion instabilities using a Helmholtz resonator with an oscillating volume. *Combust Sci Technol* 2012; 184(5): 694–716.
8. Kinsler LE., Frey AR., Coppens AB., et al. *Fundamentals of acoustics*. 4th ed. New York: John Wiley and Sons, Inc, 1999.
9. Lawn C. The acoustic impedance of perforated plates under various flow conditions relating to combustion chamber liners: Appl Acoust. Dutch 2016; 106: 144–154.
10. Vitkauskaitė G and Grubliauskas R. Perforated sound-absorbing constructions acoustic performance test and noise modeling. *Energy Procedia* 2018; 147: 288–294.
11. Howe MS, Scott MI, and Sipcic SR. the influence of tangential mean flow on the rayleigh conductivity of an aperture. *Proc Math Phys Eng Sci*, 1996, 452(1953).
12. Jing X, Sun X, Wu J, et al. Effect of grazing flow on the acoustic impedance of an orifice. *AIAA Journal* 2001; 39.
13. Peat KS, Ih JG, and Lee SH. The acoustic impedance of a circular orifice in grazing mean flow: Comparison with theory. *JASA* 2003; 114(6): 3076–3086.
14. Meng Y, Xin B, Jing X, et al. Semi-empirical impedance model of perforated plate under grazing flow[C]//In: Proceedings of the 25th AIAA/CEAS Aeroacoustics Conference. The Netherlands, May 2019: 2723.
15. Zhao D, Sun Y, Ni S, et al. Experimental and theoretical studies of aeroacoustics damping performance of a bias-flow perforated orifice. *Appl Acoust* 2019; 145: 328–338.
16. Guan D, Zhao D, Li J, et al. Aeroacoustic damping performance studies on off-axial double-layer in-duct orifices at low Mach and Helmholtz number. *Appl Acoust* 2019; 156: 46–55.
17. Guan D, Zhao D, Li J, et al. Evaluations of acoustic damping performances of double-layer in-duct perforated plates at low Mach and Helmholtz number. *JASA* 2019; 146(5): 3512–3523.
18. Yang C, Zhang P, Jacob S, et al. Investigation of extended-tube liners for control of low-frequency duct noise. *AIAA Journal* 2021; 59(10): 4179–4194.
19. Rao KN and Munjal ML. Experiment evaluation of impedance of perforates with grazing flow. *J Sound Vib* 1986; 108(2): 283–295.
20. Cummings A. The effects of grazing turbulent pipe-flow on the impedance of an orifice. *Acta Acust United Acust* 1986; 61(4): 233–242.

21. Kirby R and Cummings A. The impedance of perforated plates subjected to grazing gas flow and backed by porous media. *J Sound Vib* 1998; 217: 619–636.
22. Zhong Z and Zhao D. Time-domain characterization of the acoustic damping of a perforated liner with bias flow. *JASA* 2012; 132(1): 271–281.
23. Ji C and Zhao D. Lattice Boltzmann investigation of acoustic damping mechanism and performance of an in-duct circular orifice. *JASA* 2014; 135(6): 3243–3251.
24. Ji ZL, Xu HS, and Kang ZX. Influence of mean flow on acoustic attenuation performance of straight-through perforated tube reactive silencers and resonators. *Noise Control Eng J* 2010; 58(1): 12–17.
25. Liu C and Ji ZL. Computational Fluid Dynamics-Based Numerical Analysis of Acoustic Attenuation and Flow Resistance Characteristics of Perforated Tube Silencers. *J Vib Acoust* 2014; 136(2).
26. Zhu DD and Ji ZL. Transmission loss prediction of reactive silencers using 3-D time-domain CFD approach and plane wave decomposition technique. *Appl Acoust* 2016; 112(nov): 25–31.
27. Zhang XQ and Cheng L. Acoustic impedance of micro-perforated panels in a grazing flow. *JASA* 2019; 145(4).
28. Chen ZX, Ji ZL, and Huang HP. Acoustic impedance of perforated plates in the presence of bias flow. *J Sound Vib* 2019; 446.
29. Chen ZX, Ji ZL, and Huang H. Acoustic impedance of perforated plates in the presence of fully developed grazing flow. *J Sound Vib* 2020; 485: 115547.
30. Chu BT and Kovászny LSG. Non-linear interactions in a viscous heat-conducting compressible gas. *J Fluid Mech* 1958; 3(05): 494–514.
31. Ullrich WC, Schulze M, and Sattelmayer T. *Fundamental indirect noise generation by interactions between entropy, vorticity and acoustic waves in the context of aero engine applications*[C]//In: Proceedings of the International Congress and Exposition on Noise Control Engineering, 2015.
32. Xin B, Sun D, Jing X, et al. Numerical study of acoustic instability in a partly lined flow duct using the full linearized Navier–Stokes equations. *J Sound Vib* 2016.
33. Zhao H, Lu Z, Guan Y, et al. Effect of extended necks on transmission loss performances of Helmholtz resonators in presence of a grazing flow. *Aerosp Sci Technol* 2018; 77(jun): 228–234.
34. Huang H, Ji Z, and Han K. Acoustic Behavior Prediction and Analysis of Resonators in the Presence of Low Mach Number Flow. *J Theor Comput Acoust* 2020; 28(03): 1950015.
35. Toulorge T. *Efficient Runge-Kutta discontinuous Galerkin methods applied to aeroacoustics*[D]. PhD thesis. KU Leuven, 2012.
36. Ou Y and Zhao Y. Prediction of the Absorption Characteristics of Non-Uniform Acoustic Absorbers with Grazing Flow. *Applied Sciences* 2023; 13(4): 2256.
37. Tang Y, He W, Xin F, et al. Nonlinear sound absorption of ultralight hybrid-cored sandwich panels: MSSP. United States 2020; 135: 106428.
38. Lafont V, Méry F, and Simon F. Liner Multiphysics Coupling Between Grazing Flow, Thermal Gradients, and Sound Pressure Levels. *AIAA Journal* 2022; 60(8): 4754–4763.
39. Herring Jensen MJ, Svensson E, and Shaposhnikov K. *Using the linearized Navier-Stokes equations to model acoustic liners*[C]//In: Proceedings of the 2018 AIAA/CEAS aeroacoustics conference, Atlanta, Georgia: June 25–29, 2018, p. 3783.
40. Selamet A, Radavich PM, Dickey NS, et al. Circular concentric Helmholtz resonators. *JASA* 1997; 101(1): 41–51.
41. Zhao D. Transmission loss analysis of a parallel-coupled Helmholtz resonator network. *AIAA Journal* 2012; 50(6): 1339–1346.
42. Zhao D, Ji C, and Wang B. Geometric shapes effect of in-duct perforated orifices on aeroacoustics damping performances at low Helmholtz and Strouhal number. *JASA* 2019; 145(4): 2126–2137.